Question ID: f890dc20

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question

2, 2, 2, 3, 4, 4, 11

What is the median of the seven data values shown?

Answer

A. 2

B. 3

C. 4

D. 9

Correct Answer: B

Rationale

Choice B is correct. When a data set has an odd number of values, the median can be found by ordering the values from least to greatest and determining the value in the middle. Since the values are already presented in order from least to greatest and there are 7 values, the median is the fourth value in the list. Therefore, the median is 3.

Choice A is incorrect. This is the mode. Choice C is incorrect. This is the mean. Choice D is incorrect. This is the range.

Question ID: 52f9a246

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question

$$4, 4, 4, 4, 8, 8, 8, 13, 13$$

Which frequency table correctly represents the data listed?

Answer

A.

Number	Frequency
4	4
8	3
13	2

B.

Number	Frequency
4	4
3	8
2	13

C.

Number	Frequency
4	16
8	24
13	26

D.

Number	Frequency
16	4
24	8
26	13

Correct Answer: A

Rationale

Choice A is correct. A frequency table is a table that lists the data value and shows the number of times the data value occurs. In the data listed, the number 4 occurs four times, the number 8 occurs three times, and the number 13 occurs two times. This corresponds to the table in choice A.

Choice B is incorrect. This table has the values for number and frequency reversed.

Choice C is incorrect because the frequency values don't represent the data listed.

Choice D is incorrect. This table represents the listed number values as the frequency values.

Question ID: 8736334b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question

Data set A: 72,73,73,76,76

Data set B: 61,64,74,85,x

Data set A and data set B each contain 5 numbers. If the mean of data set A is equal to the mean of data set B, what is the value of x ?

Answer

A. 77

B. 85

C. 86

D. 95

Correct Answer: C

Rationale

Choice C is correct. The mean of a data set is found by dividing the sum of the values in the data set by the number of values in the data set. Therefore, the mean of data set A is $\frac{72+73+73+76+76}{5}$, which simplifies to 74. The mean of data set B is represented by the equation $\frac{61+64+74+85+x}{5}$, or $\frac{284+x}{5}$. It's given that the mean of data set A is equal to the mean of data set B. Therefore, the equation $74=\frac{284+x}{5}$ can be used to solve for x. Multiplying both sides of this equation by 5 yields $370=284+x$. Subtracting 284 from both sides of this equation yields $86=x$.

Choices A, B, and D are incorrect and may result from calculation errors.

Question ID: c88e0663

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question

For a school fund-raiser, 10 students sold a total of 90 boxes of cookies. Which of the following can be calculated from this information?

Answer

A. The average number of boxes sold per student

B. The median number of boxes sold per student

C. The greatest number of boxes sold by one student

D. The least number of boxes sold by one student

Correct Answer: A

Rationale

Choice A is correct. The average can be found by dividing the total number of boxes sold by the number of students, which is $\frac{90}{10} = 9$.

Choices B, C, and D are incorrect. Each results from choosing measures that require the results of individual students, which are not given.

Question ID: 4b09f783

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question

A list of 10 data values is shown.

$$6, 8, 16, 4, 17, 26, 8, 5, 5, 5$$

What is the mean of these data?

Correct Answer: 10

Rationale

The correct answer is 10. The mean of a data set is calculated by dividing the sum of the data values by the number of data values in the data set. For this data set, the mean can be calculated as $\frac{6+8+16+4+17+26+8+5+5+5}{10}$, which is equivalent to $\frac{100}{10}$, or 10.

Question ID: fa7a0164

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question

The table below shows the high and low temperatures in Houston, Texas, during a five-day period.

Temperatures in Houston, Texas (degrees Fahrenheit)

	Monday	Tuesday	Wednesday	Thursday	Friday
High temperature	73	56	62	75	81
Low temperature	49	37	41	54	63

What was the mean low temperature, in degrees Fahrenheit, during the five-day period?

Answer

A. 48.8

B. 49

C. 59

D. 59.1

Correct Answer: A

Rationale

Choice A is correct. The mean low temperature can be calculated by finding the sum of the low temperatures for all the days shown in the table, 49 + 37 + 41 + 54 + 63 = 244, and then dividing the sum by the number of days the temperature was recorded, $244 \div 5 = 48.8$.

Choice B is incorrect. This may be the result of choosing the median rather than calculating the mean. Choices C and D are incorrect and may be the result of calculation errors.

Question ID: 708590d7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question

Data set A: 1, 2, 3, 4, 5, 6, 7

Data set B: 1, 1, 2, 2, 3, 3, 4

Which of the following statements correctly compares the means of data set A and data set B?

Answer

A. The mean of each data set is 2.

B. The mean of each data set is 4.

C. The mean of data set A is less than the mean of data set B.

D. The mean of data set A is greater than the mean of data set B.

Correct Answer: D

Rationale

Choice D is correct. The mean of a data set is found by dividing the sum of the values in the data set by the number of values in the data set. Therefore, the mean of data set A is $\frac{1+2+3+4+5+6+7}{7} = \frac{28}{7}$, or 4. The mean of data set B is $\frac{1+1+2+2+3+3+4}{7} = \frac{16}{7}$, or approximately 2.2857. Therefore, the mean of data set A is greater than the mean of data set B.

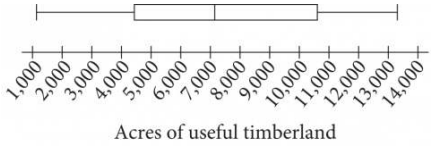
Alternate approach: Data set A and data set B are both ordered from least to greatest value. Besides the first value in each data set, which is 1, each value in ordered data set B is less than the respective value in ordered data set A. Therefore, conceptually, the mean of data set A must be greater than the mean of data set B.

Choices A, B, and C are incorrect and may result from various misconceptions or miscalculations.

Question ID: 374b18f9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question



The number of acres of useful timberland in 13 counties in California is summarized in the box plot above. Which of the following is closest to the median number of acres?

Answer

A. 4,399

B. 7,067

C. 8,831

D. 10,595

Correct Answer: B

Rationale

Choice B is correct. The median of the data summarized by a box plot is the value associated with the vertical line segment within the box. According to the box plot shown, this value is slightly greater than 7,000. Therefore, the closest value for the median number of acres is 7,067.

Choice A is incorrect. This is the value associated with the vertical line segment forming the left-hand side of the box. Choice C is incorrect. This value is greater than the value associated with the vertical line segment within the box. Choice D is incorrect. This is the value associated with the vertical line segment forming the right-hand side of the box.

Question ID: c54b92a2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question

A study was conducted on the production rates for a company that produces tractor wheels. The table below shows the number of wheels made during 11 consecutive one-hour production periods.

One-hour period	Number of wheels made
A	24
B	24
C	21
D	21
E	21
F	19
G	24
H	24
I	19
J	22
K	23

What is the range of the number of wheels made for the 11 one-hour periods?

Answer

A. 5.5

B. 5.0

C. 4.5

D. 4.0

Correct Answer: B

Rationale

Choice B is correct. Range is defined as the difference between the greatest and least values from a set of data. The greatest number of wheels made during a one-hour period was 24 wheels. The least number of wheels was 19. Hence, the range is $24 - 19 = 5$, or 5.0.

Choices A, C, and D are incorrect and may be the result of arithmetic errors or incorrectly identifying the greatest or least number of wheels made during a one-hour period.

Question ID: 66f03086

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question

$$71, 72, 73, 76, 77, 79, 83, 87, 93$$

What is the median of the data shown?

Answer

A. 71

B. 77

C. 78

D. 79

Correct Answer: B

Rationale

Choice B is correct. The median of a data set with an odd number of data values is defined as the middle value of the ordered list of values. The data set shown has nine values, so the median is the fifth value in the ordered list, which is 77.

Choice A is incorrect. This is the minimum value of the data set, not the median.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the mean of the data set, not the median.

Question ID: 4bb25495

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question

Five Smallest Countries in 2016

Country	Land area (square kilometers)
Monaco	2.0
Nauru	21
San Marino	61
Tuvalu	26
Vatican City	0.44

The table above shows the land area, in square kilometers, of the five smallest countries of the world in 2016. Based on the table, what is the mean land area of the 5 smallest countries in 2016, to the nearest square kilometer?

Answer

A. 20

B. 22

C. 61

D. 110

Correct Answer: B

Rationale

Choice B is correct. The mean land area of these 5 countries is equal to the sum of the land areas of these countries, or $2.0+21+61+26+0.44$, divided by the number of countries in the table, 5, or $\frac{2.0+21+61+26+0.44}{5}$. Combining like terms in the numerator yields $\frac{110.44}{5}$, which simplifies to 22.088 square kilometers. This value, when rounded to the nearest square kilometer, is 22.

Choice A is incorrect and may result from a calculation error. Choice C is incorrect. This is the greatest land area of the 5 countries in the table. Choice D is incorrect. This is the sum of the land areas of the 5 countries in the table, rounded to the nearest square kilometer.

Question ID: 6670e407

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question

Number of High School Students Who Completed Summer Internships

High school	Year				
	2008	2009	2010	2011	2012
Foothill	87	80	75	76	70
Valley	44	54	65	76	82
Total	131	134	140	152	152

The table above shows the number of students from two different high schools who completed summer internships in each of five years. No student attended both schools. Which of the following statements are true about the number of students who completed summer internships for the 5 years shown?

1. The mean number from Foothill High School is greater than the mean number from Valley High School.
2. The median number from Foothill High School is greater than the median number from Valley High School.

Answer

A. I only

B. II only

C. I and II

D. Neither I nor II

Correct Answer: C

Rationale

Choice C is correct. The mean of a data set is found by dividing the sum of the values by the number of values. Therefore, the mean number of students who completed summer internships from Foothill High School is $\frac{87+80+75+76+70}{5}=\frac{388}{5}$, or 77.6. Similarly, the mean number from Valley High School is $\frac{44+54+65+76+82}{5}=\frac{321}{5}$, or 64.2. Thus, the mean number from Foothill High School is greater than the mean number from Valley High School. When a data set has an odd number of elements, the median can be found by ordering the values from least to greatest and determining the value in the middle. Since there are five values in each data set, the third value in each ordered list is the median. Therefore, the median number from Foothill High School is 76 and the median number from Valley High School is 65. Thus, the median number from Foothill High School is greater than the median number from Valley High School.

Choices A, B, and D are incorrect and may result from various misconceptions or miscalculations.

Question ID: 869a32f1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question

The high temperature, in degrees Fahrenheit (°F), in a certain city was recorded for each of 5 days. The data are shown below.

Day	1	2	3	4	5
High temperature (°F)	81	80	81	81	82

Over this 5-day period, which of the following is NOT equal to 81°F?

Answer

A. Median of the high temperatures

B. Mean of the high temperatures

C. Mode of the high temperatures

D. Range of the high temperatures

Correct Answer: D

Rationale

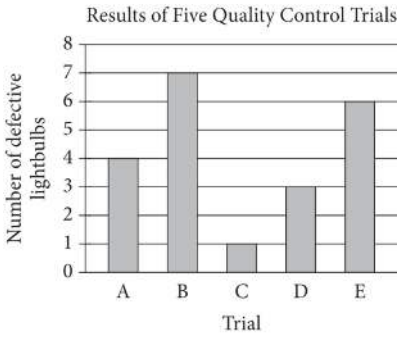
Choice D is correct. The range of a data set is the difference between the maximum and the minimum values in the set. The maximum value among the high temperatures in the table is 82°F and the minimum value is 80°F. Therefore, the range is 82°F – 80°F = 2°F.

Choice A is incorrect. The median of a data set is the middle value when the values in the set are ordered from least to greatest. Ordering the high temperatures this way gives the list 80, 81, 81, 81, 82. Therefore, the median high temperature is 81°F. Choice B is incorrect. The mean high temperature is $\frac{81+80+81+81+82}{5}=\frac{405}{5}=81$. Choice C is incorrect. The mode is the value that occurs the greatest number of times. For the set of high temperatures shown, 81 is the value that occurs 3 times, and therefore, 81°F is the mode of the high temperatures.

Question ID: a9647302

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question



For quality control, a company that manufactures lightbulbs conducted five different trials. In each trial, 500 different lightbulbs were tested. The bar graph above shows the number of defective lightbulbs found in each trial. What is the mean number of defective lightbulbs for the five trials?

Answer

A. 4.0

B. 4.2

C. 4.6

D. 5.0

Correct Answer: B

Rationale

Choice B is correct. The numbers of defective lightbulbs found for the five trials are 4, 7, 1, 3, and 6, respectively. The mean is therefore $\frac{4+7+1+3+6}{5}=4.2$.

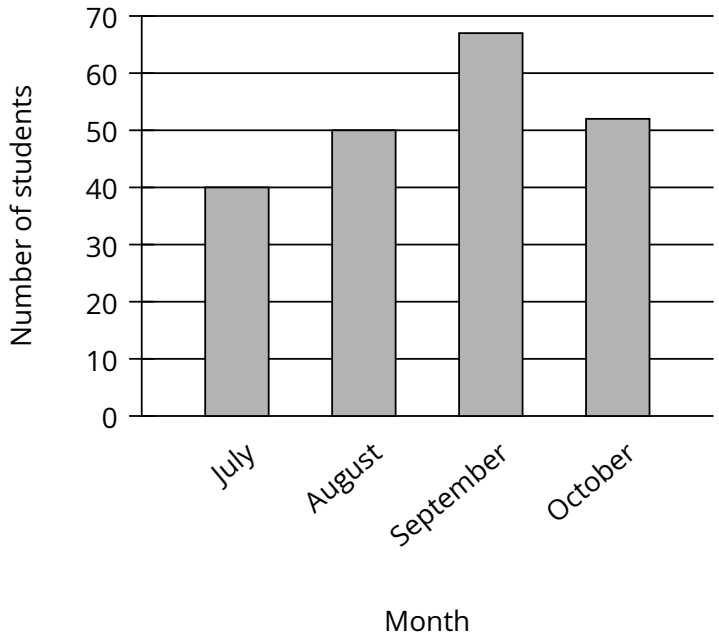
Choice A is incorrect. This is the median number of defective lightbulbs for the five trials. Choice C is incorrect and may result from an arithmetic error. Choice D is incorrect and may result from mistaking the number of trials for the number of defective lightbulbs.

Question ID: a067c926

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question

The bar graph shows the distribution of the number of students from one school who were born in one of four months.



How many more students were born in August than were born in July?

Answer

A. 90

B. 50

C. 40

D. 10

Correct Answer: D

Rationale

Choice D is correct. It's given that the bar graph shows the number of students from one school who were born in either July, August, September, or October. The bar representing the number of students born in August has a height of 50; therefore, 50 students were born in August. The bar representing the number of students born in July has a height of 40; therefore, 40 students were born in July. Thus, there were $50 - 40$, or 10 more students born in August than in July.

Choice A is incorrect. This is the total number of students born in July and August.

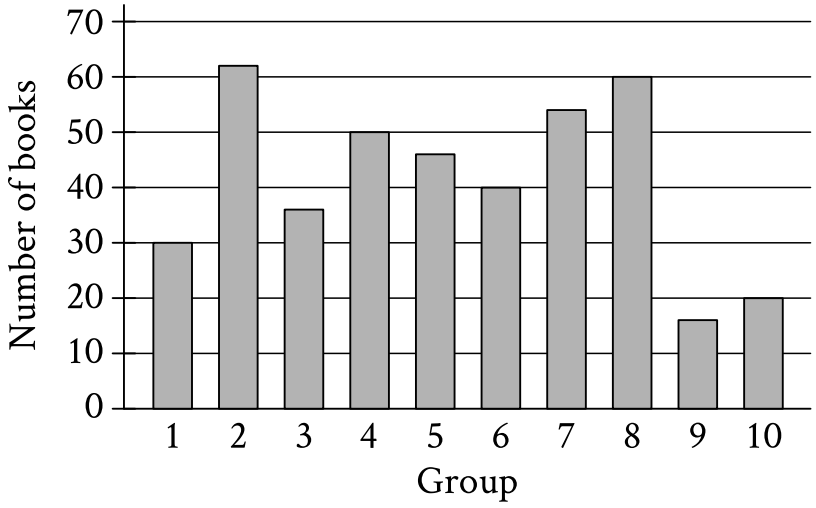
Choice B is incorrect. This is the number of students born in August.

Choice C is incorrect. This is the number of students born in July.

Question ID: 79340403

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Easy

Question



The bar graph shows the distribution of 414 books collected by 10 different groups for a book drive. How many books were collected by group 1?

Correct Answer: 30

Rationale

The correct answer is 30. The height of each bar in the bar graph shown represents the number of books collected by the group specified at the bottom of the bar. The bar for group 1 reaches a height of 30. Therefore, group 1 collected 30 books.